

SECOND-ORDER LINEAR BOUNDARY VALUE PROBLEMS AND GREEN'S FUNCTIONS

ABSTRACT. We study the two-point boundary value problem (BVP) on $[a, b]$ for the second-order linear operator

$$(Lu)(t) = a_2(t)u''(t) + a_1(t)u'(t) + a_0(t)u(t), \quad a_0, a_1, a_2 \in C([a, b]), \quad a_2(t) \neq 0.$$

Boundary conditions are given by linear functionals

$$B_i u = \sum_{j=0}^1 (a_{ij} u^{(j)}(a) + b_{ij} u^{(j)}(b)) = c_i, \quad i = 1, 2.$$

We first analyze the homogeneous ODE $Lu = 0$ and its solution space, then solve the inhomogeneous BVP $Lu = f$ via Green's functions. We provide a general matrix formulation that handles coupled boundary conditions, and then derive the classical closed form of $G(t, s)$ for separated (Dirichlet/Neumann/Robin) boundary conditions. Fredholm alternative, adjoint boundary conditions, and several worked examples are included.

CONTENTS

1. Standard form and preliminaries	2
2. The homogeneous equation $Lu = 0$	2
2.1. IVP well-posedness and solution space	2
2.2. Homogeneous BVP: matrix formulation	2
2.3. Separated boundary conditions and Wronskian	4
3. Green's function and the inhomogeneous BVP	4
3.1. Variation of parameters (scalar derivation)	4
3.2. Matrix formula for coupled boundary conditions	4
3.3. Separated boundary conditions: closed form of G	6
3.4. Enforcing inhomogeneous boundary data	7
4. Fredholm alternative and adjoint boundary conditions	7
5. Worked examples	7
6. Checks, sign conventions, and a recipe	8
Appendix: A brief derivation of the Fredholm compatibility	8
7. Appendix: Construction of fundamental matrix	9
Derivation via canonical initial value problems	9
8. Peano-Baker series: explicit integral construction of Φ	9

1. STANDARD FORM AND PRELIMINARIES

Divide by a_2 (possible since $a_2(t) \neq 0$) to put

$$u'' + p(t)u'(t) + q(t)u = r(t), \quad p = \frac{a_1}{a_2}, \quad q = \frac{a_0}{a_2}, \quad r = \frac{f}{a_2}. \quad (1)$$

We assume a_0, a_1, a_2 are continuous on $[a, b]$ and $f \in C([a, b])$. A *classical solution* is $u \in C^2([a, b])$ satisfying (1) and the given boundary conditions pointwise.

2. THE HOMOGENEOUS EQUATION $Lu = 0$

2.1. IVP well-posedness and solution space. Consider

$$u'' + p(t)u'(t) + q(t)u = 0, \quad u(t_0) = \alpha, \quad u'(t_0) = \beta.$$

By standard linear ODE theory (Picard–Lindelöf), for each $(\alpha, \beta) \in \mathbb{R}^2$ there exists a unique solution on $[a, b]$. The set of solutions is a 2-dimensional real vector space.

Let y_1, y_2 be two solutions. The *Wronskian*

$$W(t) := \begin{vmatrix} y_1(t) & y_2(t) \\ y_1'(t) & y_2'(t) \end{vmatrix} = y_1 y_2' - y_1' y_2$$

satisfies *Abel's identity*

$$W'(t) = -p(t)W(t) \quad \implies \quad W(t) = W(t_0) \exp\left(-\int_{t_0}^t p(\tau) d\tau\right). \quad (2)$$

Thus $W(t_0) \neq 0$ iff $W(t) \neq 0$ for all t , and then $\{y_1, y_2\}$ is a *fundamental pair*; every solution is $c_1 y_1 + c_2 y_2$.

Remark 2.1 (Canonical fundamental pair). A convenient choice is the pair determined by initial conditions at $t = a$:

$$y_1(a) = 1, \quad y_1'(a) = 0; \quad y_2(a) = 0, \quad y_2'(a) = 1.$$

2.2. Homogeneous BVP: matrix formulation. Write the second-order equation as a first-order system

$$x' = A(t)x, \quad x = \begin{bmatrix} u \\ u' \end{bmatrix}, \quad A(t) = \begin{bmatrix} 0 & 1 \\ -q(t) & -p(t) \end{bmatrix}.$$

Let $\Phi(t)$ be a *fundamental matrix* with $\Phi(a) = I_2$. Then $x(t) = \Phi(t)\xi$ with $\xi = x(a)$.

Encode the boundary functionals by matrices

$$U_a = \begin{bmatrix} a_{10} & a_{11} \\ a_{20} & a_{21} \end{bmatrix}, \quad U_b = \begin{bmatrix} b_{10} & b_{11} \\ b_{20} & b_{21} \end{bmatrix},$$

so that $B(u) = U_a x(a) + U_b x(b) \in \mathbb{R}^2$. For the homogeneous BVP ($B(u) = 0$), using $x(b) = \Phi(b)x(a)$ we obtain

$$S\xi = 0, \quad S := U_a + U_b \Phi(b) \in \mathbb{R}^{2 \times 2}. \quad (3)$$

Proposition 2.2 (Regularity criterion). *The only solution to $Lu = 0$ with $B(u) = 0$ is $u \equiv 0$ if and only if $\det S \neq 0$. If $\det S = 0$, then the space of homogeneous BVP solutions has dimension $\dim \ker S \in \{1, 2\}$.*

Proof. We prove the “only if” part. Suppose, to the contrary, that $\det S = 0$. Then there exists a nonzero vector $\xi \in \mathbb{R}^2$ with $S\xi = 0$. Consider the homogeneous solution $x(t) := \Phi(t)\xi$. Since $\Phi(t)$ is invertible for all t (proof below), we have $x \not\equiv 0$. Moreover,

$$B(u) = U_a x(a) + U_b x(b) = U_a \xi + U_b \Phi(b)\xi = (U_a + U_b \Phi(b))\xi = S\xi = 0.$$

Thus x (equivalently u) is a *nontrivial* solution of $Lu = 0$ satisfying the homogeneous boundary conditions, contradicting the hypothesis that the trivial solution is unique. Therefore $\det S \neq 0$. $\square$

Definition 2.3 (Adjugate / classical adjoint). Let $X = (x_{ij}) \in \mathbb{R}^{n \times n}$. For $1 \leq i, j \leq n$, let $M_{ij}(X)$ be the $(n-1) \times (n-1)$ minor obtained by deleting row i and column j from X , and let

$$C_{ij}(X) := (-1)^{i+j} \det M_{ij}(X)$$

be the (i, j) cofactor. The *adjugate* (classical adjoint) of X is

$$\text{adj}(X) := (C_{ji}(X))_{i,j=1}^n = (C_{ij}(X))^{\top}.$$

It satisfies the identity

$$\text{adj}(X) X = X \text{adj}(X) = (\det X) I_n,$$

which holds for all square X (including singular matrices).

Lemma 2.4 (Jacobi's formula). Let $X : [a, b] \rightarrow \mathbb{R}^{n \times n}$ be differentiable. Then

$$\frac{d}{dt} \det X(t) = \text{tr}(\text{adj}(X(t)) X'(t)).$$

Proof. Write the columns of $X(t)$ as $c_1(t), \dots, c_n(t)$. The determinant is multilinear and alternating in the columns, hence

$$\frac{d}{dt} \det X(t) = \sum_{i=1}^n \det(c_1(t), \dots, c_{i-1}(t), c'_i(t), c_{i+1}(t), \dots, c_n(t)).$$

Expanding the i -th term along column i gives

$$\det(\dots, c'_i(t), \dots) = \sum_{j=1}^n C_{ji}(t) x'_{ji}(t),$$

where $C_{ji}(t)$ is the cofactor of the entry $x_{ji}(t)$. Summing over i ,

$$\frac{d}{dt} \det X(t) = \sum_{i,j} C_{ji}(t) x'_{ji}(t) = \sum_{i,j} (C(t)^{\top})_{ij} X'_{ji}(t) = \sum_{i,j} \text{adj}(X(t))_{ij} X'_{ji}(t).$$

But $\sum_{i,j} \text{adj}(X)_{ij} X'_{ji} = \text{tr}(\text{adj}(X) X')$, proving the claim. $\square$

Claim. Let $\Phi'(t) = A(t)\Phi(t)$ on $[a, b]$ with $\Phi(a) = I_2$, where $A \in C([a, b]; \mathbb{R}^{2 \times 2})$. Then

$$\det \Phi(t) = \exp\left(\int_a^t \text{tr} A(\tau) d\tau\right) \neq 0 \quad \text{for all } t \in [a, b],$$

hence $\Phi(t)$ is invertible everywhere.

Proof. (Jacobi–Liouville) For any differentiable matrix $X(t)$, $\frac{d}{dt} \det X = \text{tr}(\text{adj}(X) X')$. With $X = \Phi$ and $\Phi' = A\Phi$,

$$\frac{d}{dt} \det \Phi = \text{tr}(\text{adj}(\Phi) A\Phi) = \text{tr}(A \text{adj}(\Phi) \Phi) = \text{tr}(A (\det \Phi) I_2) = (\det \Phi) \text{tr} A.$$

Solve the scalar ODE $W' = (\text{tr} A) W$ with $W(a) = \det \Phi(a) = 1$ to get $W(t) = \exp(\int_a^t \text{tr} A)$, which is never zero. $\square$

Theorem 2.5 (ODE for the inverse). Let Φ solve $\Phi' = A\Phi$, $\Phi(a) = I_2$. Then $\Phi(t)$ is invertible for all $t \in [a, b]$, and its inverse satisfies

$$\frac{d}{dt} \Phi(t)^{-1} = -\Phi(t)^{-1} A(t).$$

Proof. Differentiate $\Phi^{-1}\Phi = I$ to get $(\Phi^{-1})'\Phi + \Phi^{-1}\Phi' = 0$, hence $(\Phi^{-1})' = -\Phi^{-1}\Phi'\Phi^{-1} = -\Phi^{-1}A$. If $\Phi(t_0)$ were singular for some t_0 , the unique solution of $Y' = -YA$, $Y(a) = I$ could not reach a singular matrix at finite time; more elementarily, see the Liouville formula below showing $\det \Phi(t) \neq 0$. $\square$

2.3. Separated boundary conditions and Wronskian.

For separated BC

$$B_1 u = \alpha_1 u(a) + \beta_1 u'(a) = 0, \quad B_2 u = \alpha_2 u(b) + \beta_2 u'(b) = 0,$$

one often constructs:

- y_L : a nontrivial solution of $Lu = 0$ satisfying $B_1 y_L = 0$,
- y_R : a nontrivial solution of $Lu = 0$ satisfying $B_2 y_R = 0$.

If the homogeneous BVP is regular (unique trivial solution), then y_L, y_R are linearly independent and

$$W(t) := y_L y_R' - y_L' y_R \neq 0 \quad \text{on } [a, b].$$

3. GREEN'S FUNCTION AND THE INHOMOGENEOUS BVP

We solve

$$u'' + p u' + q u = r, \quad B(u) = c \in \mathbb{R}^2.$$

3.1. Variation of parameters (scalar derivation). Let y_1, y_2 be a fundamental pair for the homogeneous equation. Seek $u_p = v_1 y_1 + v_2 y_2$ with the auxiliary condition

$$y_1 v_1' + y_2 v_2' = 0,$$

which yields the linear system

$$\begin{cases} y_1 v_1' + y_2 v_2' = 0, \\ y_1' v_1 + y_2' v_2 = r. \end{cases}$$

Cramer's rule with $W = y_1 y_2' - y_1' y_2 \neq 0$ gives

$$v_1' = -\frac{y_2 r}{W}, \quad v_2' = \frac{y_1 r}{W}. \quad (4)$$

Integrating from a to t (constants can be absorbed into the homogeneous part),

$$u_p(t) = -y_1(t) \int_a^t \frac{y_2(s) r(s)}{W(s)} ds + y_2(t) \int_a^t \frac{y_1(s) r(s)}{W(s)} ds.$$

The general solution is $u = u_p + c_1 y_1 + c_2 y_2$; the constants c_1, c_2 are fixed by the boundary conditions.

3.2. Matrix formula for coupled boundary conditions.

Return to the system

$$x' = Ax + g, \quad g(t) = \begin{bmatrix} 0 \\ r(t) \end{bmatrix}.$$

Variation of constants yields

$$x(t) = \Phi(t) x(a) + \int_a^t \Phi(t) \Phi(s)^{-1} g(s) ds.$$

Proof. Define $c(t) := \Phi(t)^{-1} x(t)$. Since Φ is invertible on $[a, b]$, differentiate using the product rule and $\frac{d}{dt} \Phi^{-1} = -\Phi^{-1} A$:

$$c'(t) = \frac{d}{dt} (\Phi^{-1} x) = -\Phi^{-1} A x + \Phi^{-1} (Ax + g) = \Phi^{-1}(t) g(t).$$

Integrate from a to t :

$$c(t) = c(a) + \int_a^t \Phi(s)^{-1} g(s) ds = x(a) + \int_a^t \Phi(s)^{-1} g(s) ds,$$

because $c(a) = \Phi(a)^{-1} x(a) = x(a)$. Multiply by $\Phi(t)$:

$$x(t) = \Phi(t) c(t) = \Phi(t) x(a) + \Phi(t) \int_a^t \Phi(s)^{-1} g(s) ds = \Phi(t) x(a) + \int_a^t \Phi(t) \Phi(s)^{-1} g(s) ds.$$

□

Impose $B(u) = c$:

$$U_a x(a) + U_b x(b) = c \quad \Rightarrow \quad Sx(a) + U_b \int_a^b \Phi(b)\Phi(s)^{-1}g(s) ds = c,$$

with $S = U_a + U_b\Phi(b)$. If $\det S \neq 0$, then

$$x(a) = S^{-1} \left(c - U_b \int_a^b \Phi(b)\Phi(s)^{-1}g(s) ds \right),$$

and hence the first component $u(t) = e_1^\top x(t)$ satisfies

$$\begin{aligned} u(t) &= e_1^\top x(t) = e_1^\top \Phi(t) \xi + \int_a^t e_1^\top \Phi(t)\Phi(s)^{-1} e_2 \underbrace{\frac{f(s)}{a_2(s)}}_{g(s)} ds \\ &= e_1^\top \Phi(t) \xi + \int_a^t \left(e_1^\top \Phi(t)\Phi(s)^{-1} e_2 \frac{1}{a_2(s)} \right) f(s) ds. \end{aligned}$$

$$\boxed{u(t) = u_c(t) + \int_a^b G(t, s) f(s) ds,} \quad (5)$$

where

$$u_c(t) = e_1^\top \Phi(t) S^{-1} c, \quad (6)$$

$$G(t, s) = e_1^\top \left(\Phi(t)\Phi(s)^{-1} \mathbf{1}_{\{s \leq t\}} - \Phi(t)S^{-1}U_b \Phi(b)\Phi(s)^{-1} \right) e_2 \frac{1}{a_2(s)}. \quad (7)$$

Here $e_1 = (1, 0)^\top$, $e_2 = (0, 1)^\top$. This G is the Green's kernel for L with the *homogeneous* boundary conditions ($c = 0$).

Theorem 3.1 (Existence and uniqueness via Green's function). *If $\det S \neq 0$, then for every $f \in C([a, b])$ and $c \in \mathbb{R}^2$ there exists a unique $u \in C^2([a, b])$ solving $Lu = f$ with $B(u) = c$, given by (5)–(7). For each fixed s , $G(\cdot, s)$ is continuous on $[a, b]$, piecewise C^1 with jump*

$$a_2(s)(\partial_t G(s^+, s) - \partial_t G(s^-, s)) = 1,$$

and satisfies the homogeneous boundary conditions $B(G(\cdot, s)) = 0$.

Proof. Work with the first-order system

$$x' = A(t)x + g(t), \quad x = \begin{pmatrix} u \\ u' \end{pmatrix}, \quad A = \begin{bmatrix} 0 & 1 \\ -a_0/a_2 & -a_1/a_2 \end{bmatrix}, \quad g = \begin{pmatrix} 0 \\ f/a_2 \end{pmatrix}.$$

Let Φ be the canonical fundamental matrix of $x' = Ax$ ($\Phi' = A\Phi$, $\Phi(a) = I_2$), and set

$$S := U_a + U_b \Phi(b).$$

Assume $\det S \neq 0$.

Existence. Define

$$x(t) := \underbrace{\Phi(t)S^{-1}c}_{=: x_c(t)} + \int_a^b \left(\mathbf{1}_{\{s \leq t\}} \Phi(t)\Phi(s)^{-1} - \Phi(t)S^{-1}U_b \Phi(b)\Phi(s)^{-1} \right) g(s) ds.$$

(i) Differentiate under the integral (Heaviside $H(t - s)$): for a.e. t ,

$$\frac{d}{dt} \left[\Phi(t)\Phi(s)^{-1} H(t - s) \right] = A(t)\Phi(t)\Phi(s)^{-1} H(t - s) + \Phi(t)\Phi(s)^{-1} \delta(t - s).$$

Hence

$$x'(t) = A(t)x(t) + g(t),$$

so x solves the inhomogeneous system, and its first component $u = e_1^\top x$ satisfies $Lu = f$.

(ii) Boundary conditions. Evaluating x at $t = a, b$ gives

$$\begin{aligned} x(a) &= \Phi(a)S^{-1}c - \Phi(a)S^{-1}U_b \int_a^b \Phi(b)\Phi(s)^{-1}g(s) ds = S^{-1}\left(c - U_b \int_a^b \Phi(b)\Phi(s)^{-1}g\right), \\ x(b) &= \Phi(b)S^{-1}c + (I - \Phi(b)S^{-1}U_b) \int_a^b \Phi(b)\Phi(s)^{-1}g(s) ds. \end{aligned}$$

Therefore

$$\begin{aligned} U_a x(a) + U_b x(b) &= U_a S^{-1}c + U_b \Phi(b)S^{-1}c + \left[-U_a S^{-1}U_b + U_b - U_b \Phi(b)S^{-1}U_b \right] \int \dots \\ &= (U_a + U_b \Phi(b))S^{-1}c + (U_b - (U_a + U_b \Phi(b))S^{-1}U_b) \int \dots \\ &= SS^{-1}c + (U_b - SS^{-1}U_b) \int \dots = c, \end{aligned}$$

so $B(u) = c$.

Uniqueness. If u_1, u_2 are two solutions with the same (f, c) , then $w := u_1 - u_2$ satisfies $Lw = 0$ and $B(w) = 0$. In first-order form, $x'_w = Ax_w$ and $U_a x_w(a) + U_b x_w(b) = 0$. Since $\det S \neq 0$, the only homogeneous BVP solution is $x_w \equiv 0$, hence $w \equiv 0$.

Kernel properties. Fix $s \in (a, b)$ and define

$$G(t, s) := e_1^\top \left(\Phi(t)\Phi(s)^{-1} \mathbf{1}_{\{s \leq t\}} - \Phi(t)S^{-1}U_b \Phi(b)\Phi(s)^{-1} \right) e_2 \frac{1}{a_2(s)}.$$

For $t \neq s$, $G(\cdot, s)$ solves the homogeneous equation in t (each piece is a homogeneous solution), and the boundary algebra above with $c = 0$ shows $B(G(\cdot, s)) = 0$. The only singularity comes from differentiating the Heaviside factor:

$$\partial_t \left(\Phi(t)\Phi(s)^{-1} \mathbf{1}_{\{s \leq t\}} \right) = A(t)\Phi(t)\Phi(s)^{-1} \mathbf{1}_{\{s \leq t\}} + \Phi(t)\Phi(s)^{-1} \delta(t - s).$$

Taking first components and multiplying by $e_2/a_2(s)$ gives the jump

$$a_2(s)(\partial_t G(s^+, s) - \partial_t G(s^-, s)) = 1,$$

so $G(\cdot, s)$ is continuous, piecewise C^1 , with the stated jump, and yields $L_t G(\cdot, s) = \delta(\cdot - s)$ in the distributional sense. $\square$

3.3. Separated boundary conditions: closed form of G . Assume

$$B_1 u = \alpha_1 u(a) + \beta_1 u'(a) = 0, \quad B_2 u = \alpha_2 u(b) + \beta_2 u'(b) = 0.$$

Let y_L solve $Lu = 0$ with $B_1 y_L = 0$, and y_R solve $Lu = 0$ with $B_2 y_R = 0$. Define

$$W(t) := y_L(t)y'_R(t) - y'_L(t)y_R(t).$$

If the homogeneous BVP is regular, then $W(t) \neq 0$ on $[a, b]$, and the Green's function for the same (homogeneous) BC is

$$G(t, s) = \begin{cases} \frac{y_L(t)y_R(s)}{a_2(s)W(s)}, & a \leq t \leq s \leq b, \\ \frac{y_L(s)y_R(t)}{a_2(s)W(s)}, & a \leq s < t \leq b. \end{cases} \quad (8)$$

It is continuous, satisfies $B_1 G(\cdot, s) = B_2 G(\cdot, s) = 0$, and has the derivative jump $a_2(s)[G_t(s^+, s) - G_t(s^-, s)] = 1$.

Proof idea of the jump. For $t < s$, $G(\cdot, s)$ is a multiple of y_L ; for $t > s$, a multiple of y_R . The constants are chosen so that G is continuous at $t = s$ and the jump in $\partial_t G$ equals $1/a_2(s)$; a direct computation with W gives the formula. $\square$

3.4. Enforcing inhomogeneous boundary data. To solve $Lu = f$ with $B_1u = c_1$, $B_2u = c_2$, split $u = u_0 + u_f$ where

$$u_f(t) = \int_a^b G(t, s)f(s) ds \quad (\text{solves } Lu = f \text{ with homogeneous BC}),$$

and u_0 solves $Lu_0 = 0$ with $B_1u_0 = c_1$, $B_2u_0 = c_2$. If y_1, y_2 is any fundamental pair, write $u_0 = Ay_1 + By_2$ and solve

$$\begin{bmatrix} \alpha_1 y_1(a) + \beta_1 y_1'(a) & \alpha_1 y_2(a) + \beta_1 y_2'(a) \\ \alpha_2 y_1(b) + \beta_2 y_1'(b) & \alpha_2 y_2(b) + \beta_2 y_2'(b) \end{bmatrix} \begin{bmatrix} A \\ B \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}.$$

This 2×2 matrix is invertible exactly in the regular (unique) case.

4. FREDHOLM ALTERNATIVE AND ADJOINT BOUNDARY CONDITIONS

If $\det S = 0$, the homogeneous BVP has nontrivial solutions. Then the inhomogeneous BVP may fail to be solvable for some f, c . The right notion is the *Fredholm alternative*.

The *formal adjoint* of L is

$$(L^*v)(t) = (a_2v)'' - (a_1v)' + a_0v,$$

characterized by the Lagrange identity

$$\int_a^b (vLu - uL^*v) dt = \left[a_2(vu' - v'u) + (a_1 - a_2')vu \right]_{t=a}^{t=b}. \quad (9)$$

Adjoint boundary conditions $B_i^*v = 0$ are defined so that the bracket on the right of (9) vanishes whenever $B_1u = B_2u = 0$ and $B_1^*v = B_2^*v = 0$.

Theorem 4.1 (Fredholm alternative). *Suppose the homogeneous BVP ($Lu = 0$, $B(u) = 0$) has nontrivial solutions. Then the inhomogeneous BVP $Lu = f$ with homogeneous BC ($c = 0$) is solvable if and only if*

$$\int_a^b v(t)f(t) dt = 0 \quad \text{for all } v \in \ker(L^*, B^*).$$

When solvable, the set of solutions is an affine line $u_p + \ker(L, B)$.

Remark 4.2 (Separated Robin adjoint). If $B_1u = \alpha_1u(a) + \beta_1u'(a) = 0$ and $B_2u = \alpha_2u(b) + \beta_2u'(b) = 0$, one convenient choice of adjoint BC is

$$\alpha_1(a_2v' - (a_1 - a_2')v)(a) - \beta_1(a_2v)(a) = 0, \quad \alpha_2(a_2v' - (a_1 - a_2')v)(b) - \beta_2(a_2v)(b) = 0,$$

which indeed annihilates the boundary bracket in (9).

5. WORKED EXAMPLES

Example 5.1 (Dirichlet, $Lu = u''$). Let $u'' = f$ on $[a, b]$, $u(a) = u(b) = 0$. Choose $y_L(t) = t - a$, $y_R(t) = b - t$; then $W(t) = -(b - a)$ and

$$G(t, s) = \begin{cases} -\frac{(t-a)(b-s)}{b-a}, & t \leq s, \\ -\frac{(s-a)(b-t)}{b-a}, & t \geq s. \end{cases}$$

Hence $u(t) = \int_a^b G(t, s)f(s) ds$. For $f \equiv 1$ this yields $u(t) = \frac{(t-a)(b-t)}{2}$.

Example 5.2 (Dirichlet, $Lu = u'' - k^2u$ with $k > 0$). Homogeneous solutions are $\sinh/\cosh$. Pick $y_L(t) = \sinh k(t - a)$, $y_R(t) = \sinh k(b - t)$. Then $W(t) = -k \sinh k(b - a)$ and

$$G(t, s) = \begin{cases} -\frac{\sinh k(t-a) \sinh k(b-s)}{k \sinh k(b-a)}, & t \leq s, \\ -\frac{\sinh k(s-a) \sinh k(b-t)}{k \sinh k(b-a)}, & t \geq s. \end{cases}$$

Example 5.3 (Robin–Dirichlet, $Lu = u''$). On $[a, b]$ impose $u'(a) + \alpha u(a) = 0$ and $u(b) = 0$. Homogeneous solutions of $u'' = 0$ are $A + Bt$. Left BC gives $B = -\alpha A$, so $y_L(t) = 1 - \alpha(t - a)$. Right BC gives $y_R(t) = b - t$. Then $W(t) = -1 + \alpha(b - a)$ (constant). For $\alpha(b - a) \neq 1$,

$$G(t, s) = \begin{cases} \frac{[1 - \alpha(t - a)](b - s)}{-1 + \alpha(b - a)}, & t \leq s, \\ \frac{[1 - \alpha(s - a)](b - t)}{-1 + \alpha(b - a)}, & t \geq s. \end{cases}$$

If $\alpha(b - a) = 1$ we are at resonance (nontrivial homogeneous solution meets both BC); apply the Fredholm alternative.

Example 5.4 (Coupled BC via the matrix method). Let $Lu = u''$ on $[0, 1]$ with periodic BC $u(0) = u(1)$, $u'(0) = u'(1)$. Then

$$U_a = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad U_b = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}, \quad \Phi(1) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

Hence $S = U_a + U_b\Phi(1) = \begin{bmatrix} 0 & -1 \\ 0 & 0 \end{bmatrix}$ is singular: constants solve the homogeneous BVP. The inhomogeneous periodic BVP $u'' = f$ is solvable iff $\int_0^1 f(s) ds = 0$ (orthogonality to constants, the adjoint kernel).

6. CHECKS, SIGN CONVENTIONS, AND A RECIPE

- **Jump condition:** For $Lu = a_2u'' + a_1u' + a_0u$, the Green kernel satisfies

$$a_2(s)(G_t(s^+, s) - G_t(s^-, s)) = 1.$$

- **Sign of G :** With $L = u''$ and Dirichlet BC, $G \leq 0$; with $L = -u''$, $G \geq 0$.
- **Wronskian:** If nonzero at one point, W never vanishes on $[a, b]$ (Abel).
- **Quick recipe:** Put in standard form; build y_L, y_R for separated BC and use (8); or compute $\Phi(b)$, form $S = U_a + U_b\Phi(b)$ and use (7) for coupled BC. Assemble $u = u_c + \int Gf$ with u_c from (6).

APPENDIX: A BRIEF DERIVATION OF THE FREDHOLM COMPATIBILITY

For homogeneous BC ($c = 0$) and singular S , the linear equation

$$S\xi = -U_b \int_a^b \Phi(b)\Phi(s)^{-1}g(s) ds$$

has a solution iff the right-hand side is orthogonal to every $w \in \ker S^\top$:

$$w^\top U_b \int_a^b \Phi(b)\Phi(s)^{-1}g(s) ds = 0.$$

Define $v(s) := \frac{1}{a_2(s)}w^\top U_b\Phi(b)\Phi(s)^{-1}e_2$. One checks $L^*v = 0$ and $B^*v = 0$. Then

$$w^\top U_b\Phi(b)\Phi(s)^{-1}g(s) = v(s)f(s),$$

so solvability is equivalent to $\int_a^b v(s)f(s) ds = 0$ for all $v \in \ker(L^*, B^*)$.

Further reading. W. Walter, *Ordinary Differential Equations*; E. Zauderer, *Partial Differential Equations of Applied Mathematics* (Green's functions ch.); classical Sturm–Liouville notes for the self-adjoint case.

7. APPENDIX: CONSTRUCTION OF FUNDAMENTAL MATRIX

Definition 7.1 (Fundamental matrix). A matrix function $\Phi : [a, b] \rightarrow \mathbb{R}^{2 \times 2}$ is a *fundamental matrix* for $x' = A(t)x$ if

$$\Phi'(t) = A(t)\Phi(t) \quad \text{for all } t \in [a, b], \quad \det \Phi(t) \neq 0 \text{ for all } t.$$

If we additionally require $\Phi(a) = I_2$, we call it the *canonical* fundamental matrix at $t = a$.

Derivation via canonical initial value problems. By basic linear ODE theory, for any initial vector $\xi \in \mathbb{R}^2$ there is a unique solution $x(\cdot; \xi)$ of

$$x'(t) = A(t)x(t), \quad x(a) = \xi.$$

Let $e_1 = (1, 0)^\top$, $e_2 = (0, 1)^\top$, and define the two canonical homogeneous solutions

$$x^{(1)}(t) := x(t; e_1), \quad x^{(2)}(t) := x(t; e_2).$$

Set

$$\Phi(t) := \begin{bmatrix} x^{(1)}(t) & x^{(2)}(t) \end{bmatrix}. \quad (10)$$

Then:

- $\Phi(a) = [e_1 \ e_2] = I_2$ by construction.
- Differentiating columnwise gives $\Phi'(t) = \begin{bmatrix} A(t)x^{(1)}(t) & A(t)x^{(2)}(t) \end{bmatrix} = A(t)\Phi(t)$.
- For any initial vector $\xi \in \mathbb{R}^2$, linearity and uniqueness imply

$$x(t) = \Phi(t)\xi \quad \text{is the unique solution with } x(a) = \xi. \quad (11)$$

Thus Φ spans the solution space at each t and has nonzero determinant.

Theorem 7.2 (Canonical fundamental matrix). *With $A \in C([a, b]; \mathbb{R}^{2 \times 2})$, the matrix Φ defined by (10) is the unique solution of*

$$\Phi'(t) = A(t)\Phi(t), \quad \Phi(a) = I_2.$$

Moreover, every homogeneous solution is of the form $x(t) = \Phi(t)\xi$ for a unique $\xi \in \mathbb{R}^2$.

Remark 7.3 (Scalar second order view). If y_1, y_2 form a fundamental pair for the scalar homogeneous ODE $u'' + pu' + qu = 0$, then

$$\Phi(t) = \begin{bmatrix} y_1(t) & y_2(t) \\ y_1'(t) & y_2'(t) \end{bmatrix}$$

also satisfies $\Phi' = A\Phi$. Choosing y_1, y_2 via the canonical IVPs at $t = a$ gives $\Phi(a) = I_2$.

8. PEANO–BAKER SERIES: EXPLICIT INTEGRAL CONSTRUCTION OF Φ

Another way to *derive* Φ is to integrate the homogeneous system:

$$x(t) = x(a) + \int_a^t A(\tau)x(\tau) d\tau.$$

Iterating this Volterra equation yields the Peano–Baker series. Define

$$\begin{aligned} \Phi_0(t) &:= I_2, \\ \Phi_1(t) &:= \int_a^t A(\tau_1) d\tau_1, \\ \Phi_2(t) &:= \int_a^t A(\tau_1) \int_a^{\tau_1} A(\tau_2) d\tau_2 d\tau_1, \\ &\vdots \\ \Phi_n(t) &:= \int_a^t A(\tau_1) \int_a^{\tau_1} A(\tau_2) \cdots \int_a^{\tau_{n-1}} A(\tau_n) d\tau_n \cdots d\tau_1. \end{aligned}$$

Set the series

$$\Phi(t) := \sum_{n=0}^{\infty} \Phi_n(t). \quad (12)$$

Because A is continuous on a compact interval, there is M with $\|A(\tau)\| \leq M$. Then

$$\|\Phi_n(t)\| \leq \frac{(M|t-a|)^n}{n!},$$

so (12) converges absolutely and uniformly on $[a, b]$. Termwise differentiation is justified, giving

$$\Phi'(t) = A(t) \sum_{n=0}^{\infty} \Phi_n(t) = A(t)\Phi(t), \quad \Phi(a) = I_2.$$

Thus the Peano–Baker series produces the same canonical fundamental matrix.

Remark 8.1 (Time-ordered exponential). The series (12) is sometimes denoted $\Phi(t) = \mathcal{T} \exp(\int_a^t A(\tau) d\tau)$, where $\mathcal{T}$ indicates the ordering $a \leq \tau_n \leq \cdots \leq \tau_1 \leq t$.